Summary Statistics of HMDA

2026-10-13

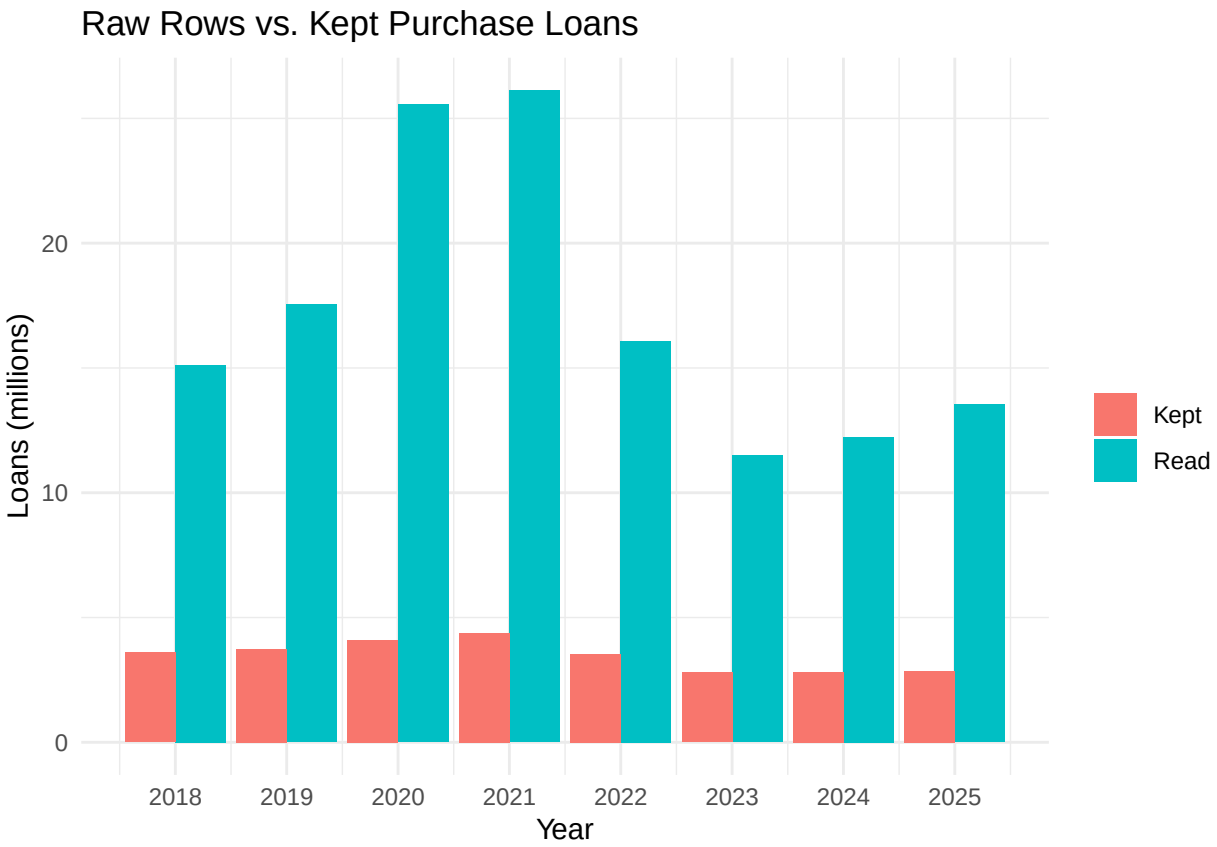
Rows read vs. kept

```
log <- read_csv(here("outputs", "hmda_cleaning_log.csv"))

log %>%
  mutate(year = as.character(year)) %>%
  select(Year = year, `Rows read` = rows_read, `Rows kept` = rows_kept) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Year	Rows read	Rows kept
2018	15,119,651	3,591,478
2019	17,545,457	3,732,655
2020	25,551,868	4,098,299
2021	26,124,552	4,369,605
2022	16,080,210	3,529,813
2023	11,483,889	2,795,089
2024	12,229,298	2,819,843
2025	13,543,606	2,852,555

```
log %>%
  pivot_longer(c(rows_read, rows_kept), names_to = "type", values_to = "rows") %>%
  mutate(type = recode(type, rows_read = "Read", rows_kept = "Kept")) %>%
  ggplot(aes(x = year, y = rows / 1e6, fill = type)) +
    geom_col(position = "dodge") +
    scale_x_continuous(breaks = 2018:2025) +
    labs(x = "Year", y = "Loans (millions)", fill = "", title = "Raw Rows vs. Kept Purchase Loans") +
    theme_minimal()
```



Loan-level summary stats by year

```
summarise_year <- function(year) {
  df <- read_dta(
    here("data", "clean", "hmda", paste0("hmda_clean_", year, ".dta")),
    col_select = c(loan_type, loan_amount, income, combined_loan_to_value_ratio,
                   interest_rate, property_value)
  )
  tibble(
    year = as.character(year),
    N = nrow(df),
    mean_loan = mean(df$loan_amount, na.rm = TRUE),
    median_loan = median(df$loan_amount, na.rm = TRUE),
    mean_income = mean(df$income, na.rm = TRUE),
    mean_rate = mean(df$interest_rate, na.rm = TRUE),
    mean_cltv = mean(df$combined_loan_to_value_ratio, na.rm = TRUE),
    share_fha = mean(df$loan_type == 2),
    share_va = mean(df$loan_type == 3),
    miss_cltv = mean(is.na(df$combined_loan_to_value_ratio)),
    miss_rate = mean(is.na(df$interest_rate)),
    miss_value = mean(is.na(df$property_value))
  )
}

by_year <- map_dfr(2018:2025, summarise_year)
```

```r
by_year %>%
  select(Year = year, N, `Mean Loan` = mean_loan, `Median Loan` = median_loan,
         `Mean Rate` = mean_rate, `Mean CLTV` = mean_cltv) %>%
  knitr::kable(digits = 1, format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

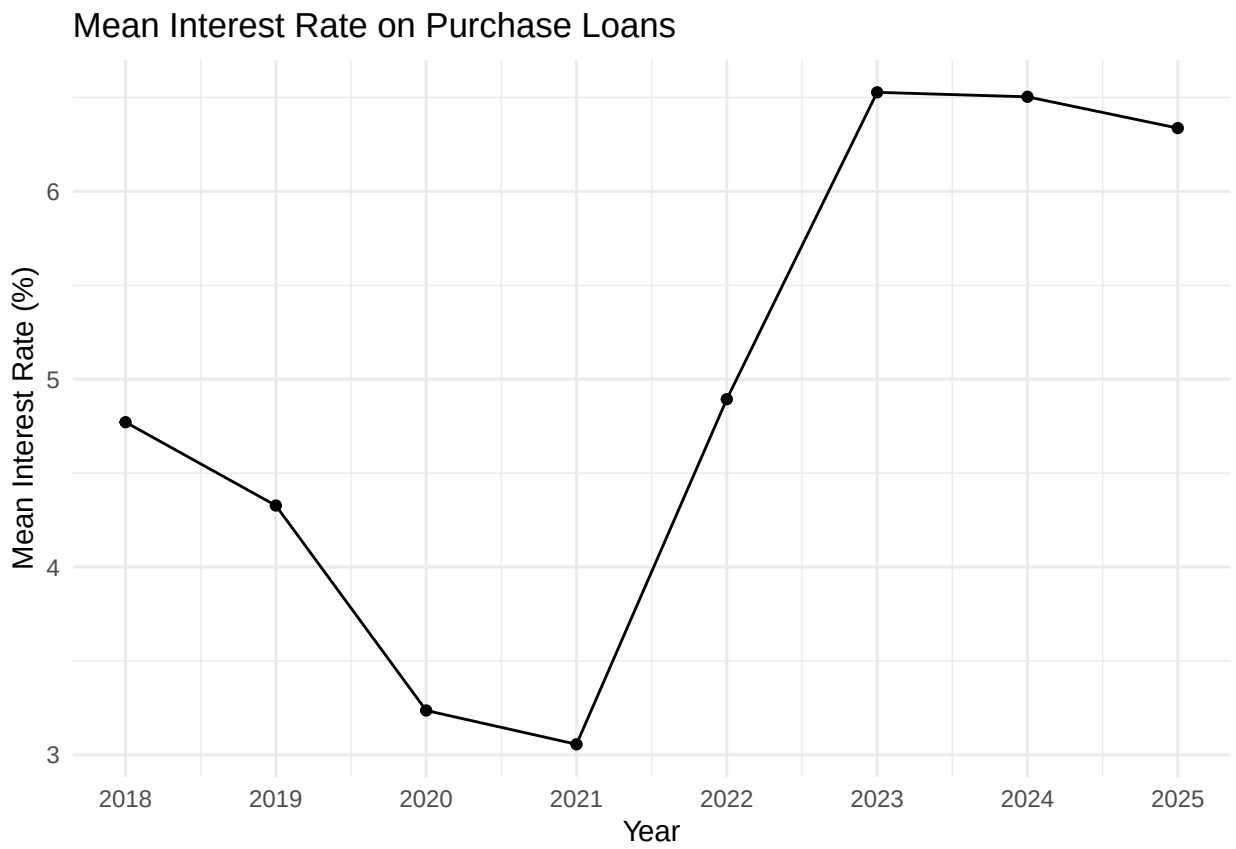
Year	N	Mean Loan	Median Loan	Mean Rate	Mean CLTV
2018	3,591,478	274,591.1	225,000	4.8	102.3
2019	3,732,655	286,242.0	235,000	4.3	92.4
2020	4,098,299	308,040.4	255,000	3.2	88.7
2021	4,369,605	352,672.5	285,000	3.1	86.4
2022	3,529,813	381,035.5	315,000	4.9	85.0
2023	2,795,089	375,694.5	315,000	6.5	84.7
2024	2,819,843	392,808.6	325,000	6.5	84.6
2025	2,852,555	403,847.3	335,000	6.3	84.8

Loan type shares and missing shares (`loan_type`: 1 = conventional, 2 = FHA, 3 = VA, 4 = USDA).

```r
by_year %>%
  select(Year = year, `FHA` = share_fha, `VA` = share_va,
         `No CLTV` = miss_cltv, `No Rate` = miss_rate, `No Value` = miss_value) %>%
  mutate(across(-Year, ~ scales::percent(.x, accuracy = 0.1))) %>%
  knitr::kable() %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Year	FHA	VA	No CLTV	No Rate	No Value
2018	19.8%	10.3%	5.5%	3.2%	3.1%
2019	20.1%	10.6%	6.0%	2.6%	3.1%
2020	19.4%	10.4%	4.7%	2.2%	2.5%
2021	17.2%	9.6%	3.8%	1.4%	1.7%
2022	16.3%	10.2%	4.3%	1.6%	1.9%
2023	19.8%	10.6%	4.5%	1.9%	2.2%
2024	20.4%	10.8%	4.7%	2.1%	2.4%
2025	21.2%	11.3%	5.3%	2.3%	2.7%

```r
by_year %>%
  mutate(year = as.numeric(year)) %>%
  ggplot(aes(x = year, y = mean_rate)) +
    geom_line() +
    geom_point() +
    scale_x_continuous(breaks = 2018:2025) +
    labs(x = "Year", y = "Mean Interest Rate (%)", title = "Mean Interest Rate on Purchase Loans") +
    theme_minimal()
```

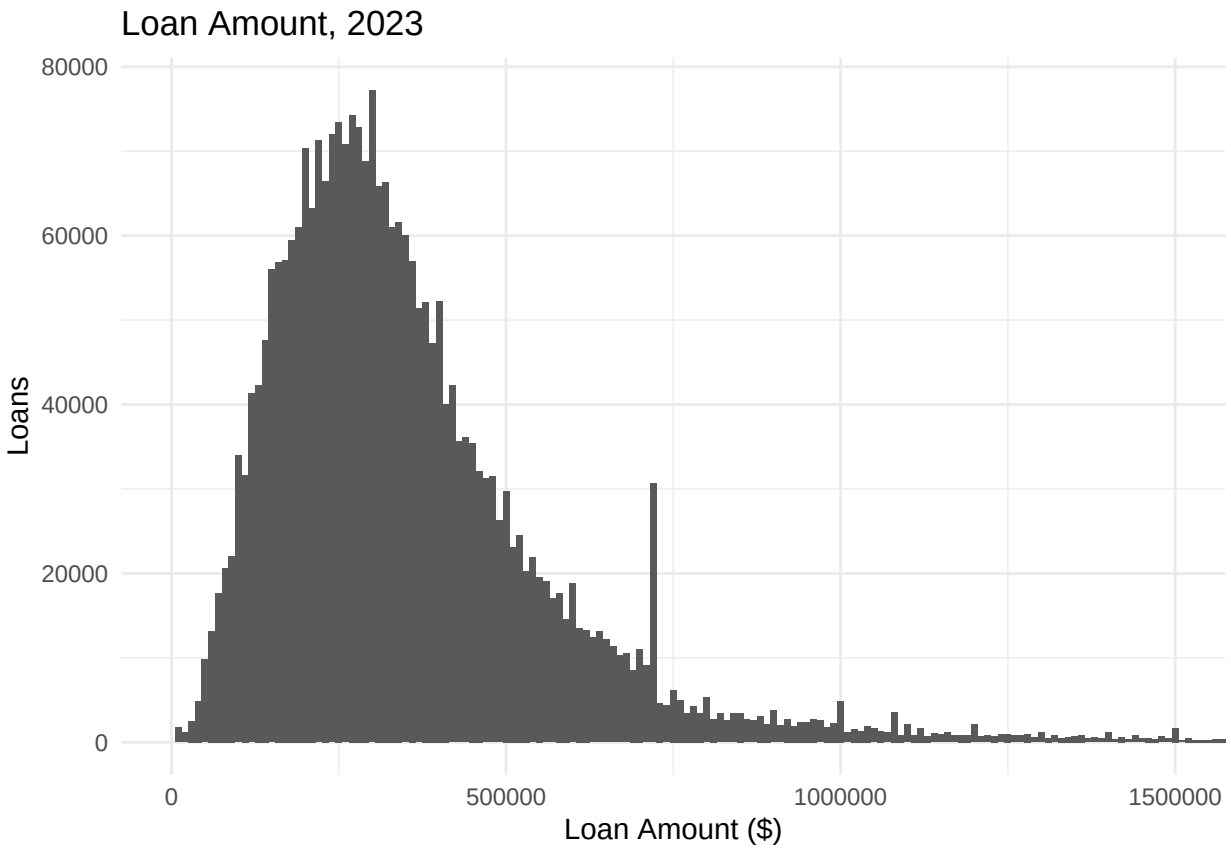


Distributions (2023)

One year only, the plots are actually readable.

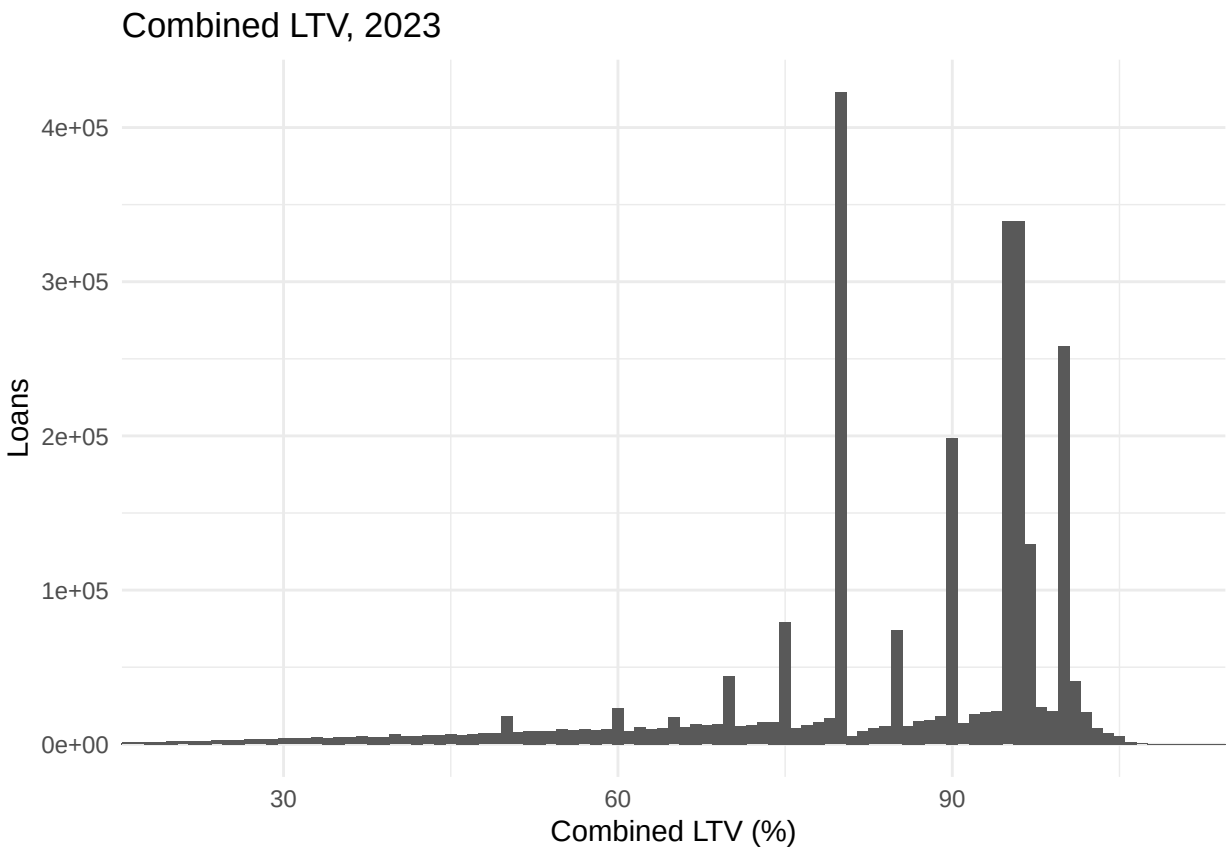
```
df23 <- read_dta(
  here("data", "clean", "hmda", "hmda_clean_2023.dta"),
  col_select = c(loan_amount, combined_loan_to_value_ratio, interest_rate)
)

ggplot(df23, aes(x = loan_amount)) +
  geom_histogram(binwidth = 10000) +
  coord_cartesian(xlim = c(0, 1500000)) +
  labs(x = "Loan Amount ($)", y = "Loans", title = "Loan Amount, 2023") +
  theme_minimal()
```

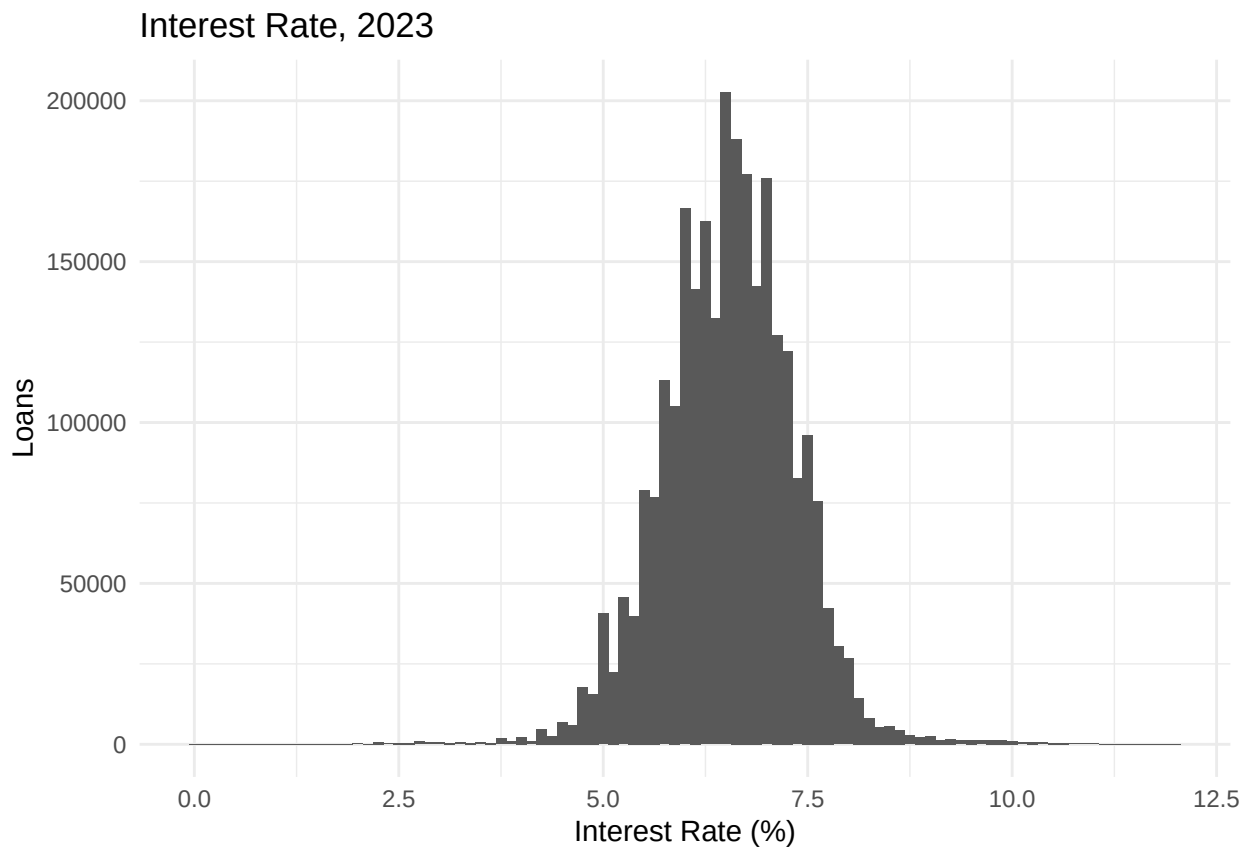


Combined loan-to-value ratio. The spikes at 80 and 96.5 are the usual 20%-down and FHA minimum-down-payment cutoffs.

```
df23 %>%
  filter(!is.na(combined_loan_to_value_ratio)) %>%
  ggplot(aes(x = combined_loan_to_value_ratio)) +
    geom_histogram(binwidth = 1) +
    coord_cartesian(xlim = c(20, 110)) +
    labs(x = "Combined LTV (%)", y = "Loans", title = "Combined LTV, 2023") +
    theme_minimal()
```




```
df23 %>%
  filter(!is.na(interest_rate), interest_rate > 0, interest_rate < 12) %>%
  ggplot(aes(x = interest_rate)) +
    geom_histogram(binwidth = 0.125) +
    labs(x = "Interest Rate (%)", y = "Loans", title = "Interest Rate, 2023") +
    theme_minimal()
```




```
rm(df23)
```

Tract-year file

```
tracts <- read_dta(here("data", "clean", "hmda", "hmda_tract_year.dta"))

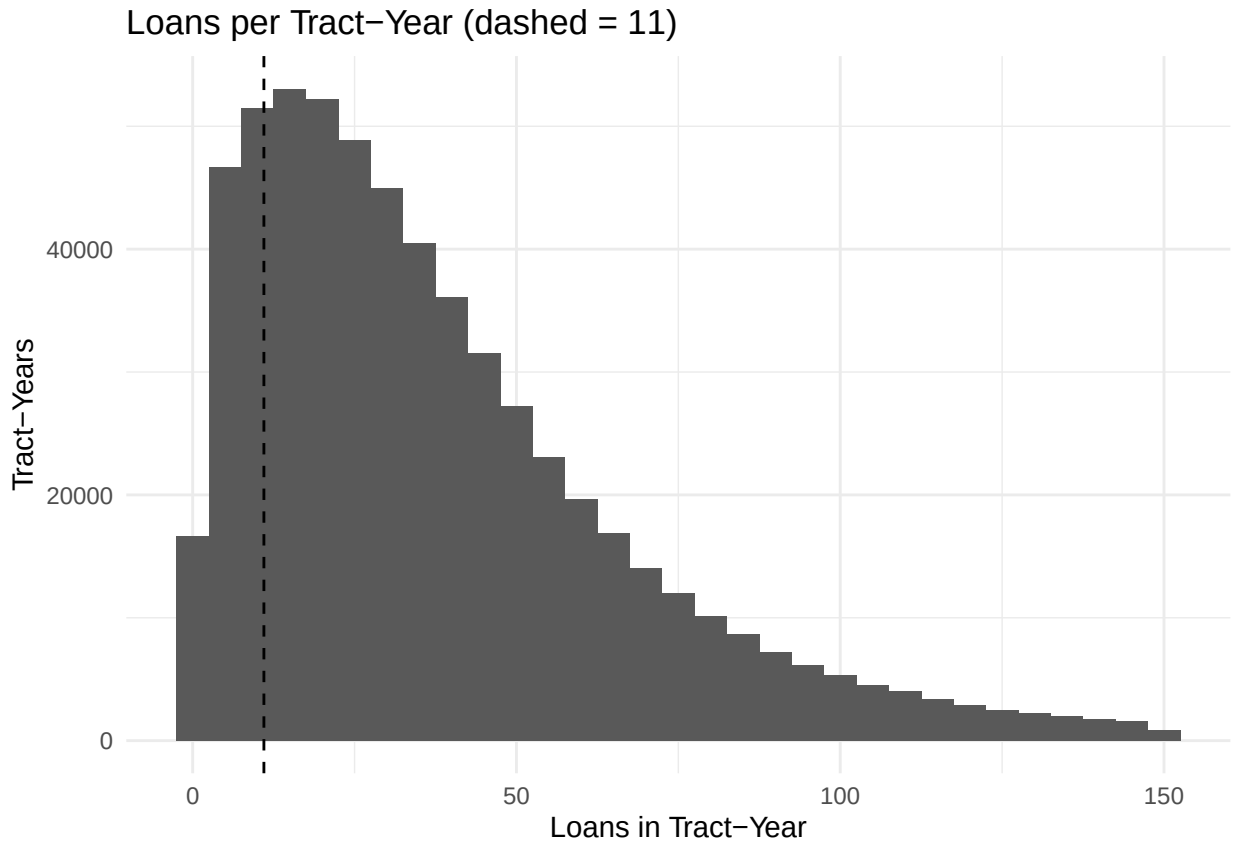
tracts %>%
  group_by(activity_year) %>%
  summarise(
    Tracts = n(),
    `Median Loans` = median(n_loans),
    `Share < 11 Loans` = scales::percent(mean(n_loans < 11), accuracy = 0.1)
  ) %>%
  rename(Year = activity_year) %>%
  mutate(Year = as.character(Year)) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Year	Tracts	Median Loans	Share < 11 Loans
2018	71,512	37	14.6%
2019	71,592	38	13.8%
2020	71,615	41	13.1%
2021	71,899	45	10.6%
2022	82,808	33	13.6%

2023	82,586	25	18.3%
2024	82,545	25	18.3%
2025	82,543	26	18.3%

The number of tracts jumps in 2022 because HMDA switches from 2010 to 2020 census tract boundaries.

```
tracts %>%
  filter(n_loans <= 150) %>%
  ggplot(aes(x = n_loans)) +
    geom_histogram(binwidth = 5) +
    geom_vline(xintercept = 11, linetype = "dashed") +
    labs(x = "Loans in Tract-Year", y = "Tract-Years",
         title = "Loans per Tract-Year (dashed = 11)") +
    theme_minimal()
```




```
tracts %>%
  mutate(state_fips = substr(census_tract, 1, 2)) %>%
  group_by(state_fips) %>%
  summarise(count = sum(n_loans)) %>%
  arrange(desc(count)) %>%
  head(10) %>%
  mutate(state_name = fips_info(state_fips)$full) %>%
  select(state_fips, state_name, count) %>%
  knitr::kable(format.args = list(big.mark = ","))
```

state_fips	state_name	count
48	Texas	2,653,530
06	California	2,220,077
12	Florida	2,210,174
37	North Carolina	1,067,558
17	Illinois	1,051,620
13	Georgia	1,036,358
39	Ohio	1,020,298
36	New York	950,856
42	Pennsylvania	941,023
51	Virginia	865,417

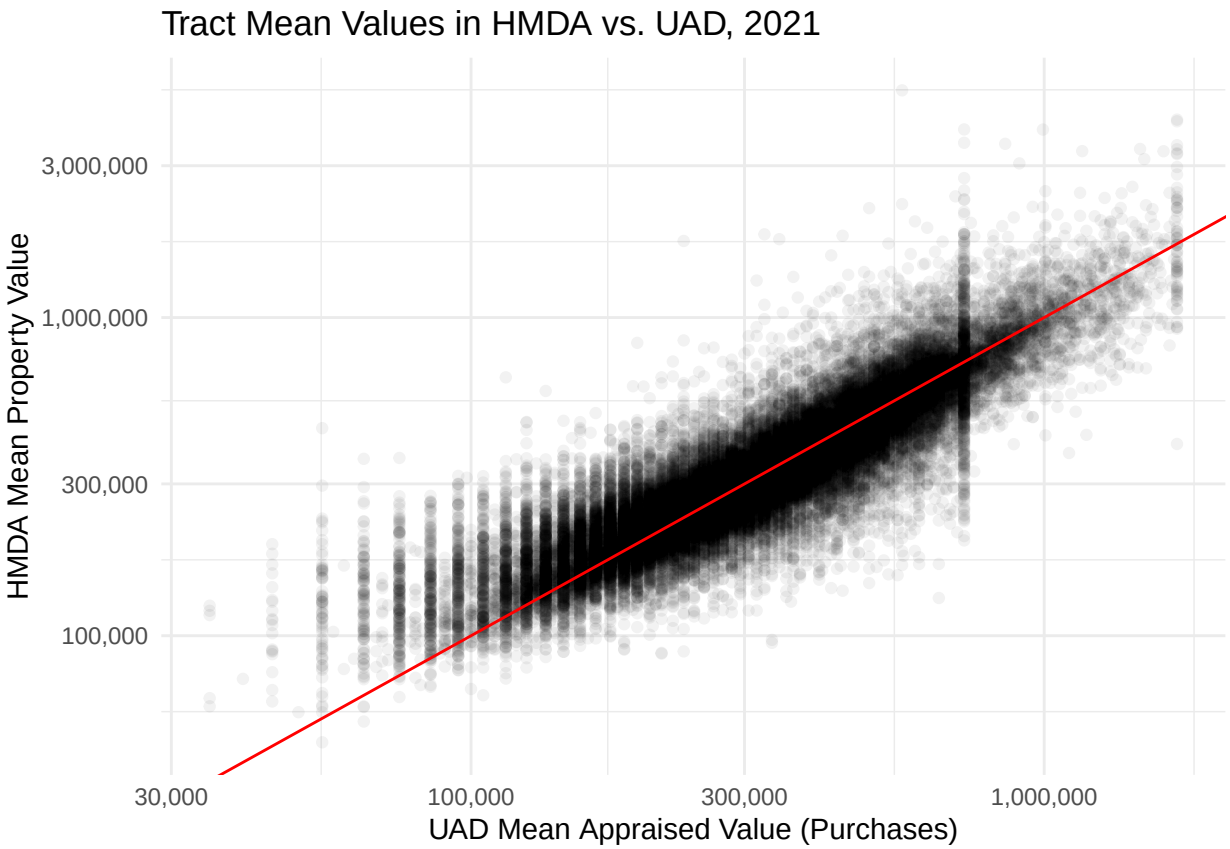
Linked HMDA-UAD file

```
linked <- read_dta(here("data", "combined", "hmda_uad_tract_year.dta"))

linked %>%
  count(year, link_status) %>%
  pivot_wider(names_from = link_status, values_from = n, values_fill = 0) %>%
  mutate(`Share HMDA Matched` = scales::percent(both / (both + hmda_only), accuracy = 0.1),
         year = as.character(year)) %>%
  rename(Year = year, Both = both, `HMDA Only` = hmda_only, `UAD Only` = uad_only) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Year	Both	HMDA Only	UAD Only	Share HMDA Matched
2018	45,635	25,877	2	63.8%
2019	48,802	22,790	2	68.2%
2020	54,119	17,496	11	75.6%
2021	54,435	17,464	10	75.7%
2022	49,146	33,662	1	59.3%

```
linked %>%
  filter(link_status == "both", year == 2021,
         !is.na(hmda_property_value_mean), !is.na(uad_purchase_value_mean)) %>%
  ggplot(aes(x = uad_purchase_value_mean, y = hmda_property_value_mean)) +
    geom_point(alpha = 0.05) +
    geom_abline(slope = 1, intercept = 0, color = "red") +
    scale_x_log10(labels = scales::comma) +
    scale_y_log10(labels = scales::comma) +
    labs(x = "UAD Mean Appraised Value (Purchases)", y = "HMDA Mean Property Value",
         title = "Tract Mean Values in HMDA vs. UAD, 2021") +
    theme_minimal()
```


Tract Mean Values in HMDA vs. UAD, 2021
HMDA Mean Property Value
3,000,000
1,000,000
300,000
100,000
30,000
100,000
300,000
1,000,000
UAD Mean Appraised Value (Purchases)